COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10-5-26	**DATE SUBMITTED:**	10-6-26
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ACTIVITY NO. 10 **Install, Configure, and Manage Log Monitoring tools**

1. PROCEDURE OUTPUT

1. Create a playbook that: a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
5. Make sure to create a new repository in GitHub for this activity.

1) Ansible tree/structure

```
masongsong@Masongsong:~/CPE212_Activity10$ tree
.
├── ansible.cfg
├── inventory.ini
├── roles
│   ├── elasticsearch
│   │   └── tasks
│   │       └── main.yml
│   ├── kibana
│   │   └── tasks
│   │       └── main.yml
│   └── logstash
│       └── tasks
│           └── main.yml
└── site.yml
```

2) Ansible.cfg

```
  GNU nano 7.2                                  ansible.cfg
[defaults]
inventory = inventory.ini
interpreter_python = auto_silent
deprecation_warnings = False
host_key_checking = False
```

3) Inventory.ini **(masong = centos, masongsong = ubuntu)**

```
  GNU nano 7.2                                inventory.ini *
[elasticsearch]
192.168.56.104 ansible_user=masongsong

[kibana]
192.168.56.103 ansible_user=masongsong

[logstash]
192.168.56.102 ansible_user=masong
```

4) Site.yml

```
  GNU nano 7.2                          site.yml
---
- name: Install and configure Elasticsearch
  hosts: elasticsearch
  become: true
  roles:
    - elasticsearch

  tasks:
    - name: Wait for Elasticsearch to become available
      wait_for:
        host: 192.168.56.104
        port: 9200
        delay: 5
        timeout: 120

- name: Install and configure Kibana
  hosts: kibana
  become: true
  roles:
    - kibana
```

```
tasks:
  - name: Wait for Kibana to become available
    wait_for:
      host: 192.168.56.103
      port: 5601
      delay: 5
      timeout: 180

- name: Install and configure Logstash
  hosts: logstash
  become: true
  roles:
    - logstash
```

5) Build main.yml for elasticsearch

```
 GNU nano 7.2              roles/elasticsearch/tasks/main.yml
---
- name: Install required dependencies
  apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
      - curl
    state: present
    update_cache: true

- name: Download Elastic GPG key
  shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearm>
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

- name: Add Elastic 8.x repository
  apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https:>
```

```
    filename: elastic-8.x
    state: present

- name: Install Elasticsearch
  apt:
    name: elasticsearch
    state: present
    update_cache: true

- name: Deploy clean Elasticsearch configuration
  copy:
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'
    content: |
      cluster.name: my-application
      node.name: node-1
      path.data: /var/lib/elasticsearch
      path.logs: /var/log/elasticsearch
      network.host: 0.0.0.0
      http.port: 9200
      discovery.type: single-node
      xpack.security.enabled: false
      xpack.security.enrollment.enabled: false
      xpack.security.http.ssl.enabled: false
      xpack.security.transport.ssl.enabled: false

- name: Reload systemd units
  systemd:
    daemon_reload: true

- name: Enable and start Elasticsearch
  systemd:
    name: elasticsearch
    enabled: true
    state: restarted
```

Analysis:

- **This main.yml for elasticsearch installs all the required dependencies, elastic's GPG key, and adds elastic 8.x apt repository so the package can be trusted and installed. There are also lines that configure elasticsearch to be able to listen on 0.0.0.0:9200. After that, the systemd reloads, enables, and restarts the service to ensure that elastic is able to function correctly.**

6) Build main.yml for kibana

```
  GNU nano 7.2                  roles/kibana/tasks/main.yml
---
- name: Install required packages
  apt:
    name:
      - apt-transport-https
      - wget
      - gnupg
      - curl
    state: present
    update_cache: true

- name: Download Elastic GPG key
  shell: |
    wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearm>
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

- name: Add Elastic 8.x repository
  apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https:>
                              [ Read 47 lines ]
    filename: elastic-8.x
    state: present

- name: Install Kibana
  apt:
    name: kibana
    state: present
    update_cache: true

- name: Configure Kibana network host
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?server.host:'
    line: 'server.host: "0.0.0.0"'
```

```
- name: Configure Kibana Elasticsearch host URL
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?elasticsearch.hosts:'
    line: 'elasticsearch.hosts: ["http://192.168.56.104:9200"]'

- name: Enable and start Kibana
  systemd:
    name: kibana
    enabled: true
    state: restarted
    daemon_reload: true
```

Analysis:

- **This main.yml for kibana adds almost the same as elasticsearch, with its GPG key and 8.x repository, then it installs kibana with the use of lineinfile in order to edit kibana.yml. It also sets the server host to 0.0.0.0 so the dashboard will be reachable on port 5601 and its elasticsearch.hosts to the elasticsearch server at 192.168.56.104(ubuntu VM). Lastly this ends with it enabling and restarting kibana to apply the changes made.**

7) Build main.yml for logstash

```
  GNU nano 7.2                roles/logstash/tasks/main.yml
---
- name: Import Elastic GPG key
  rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present

- name: Add Elastic 8.x repository
  yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true

- name: Install Logstash
  dnf:
    name: logstash
    state: present
```

```
- name: Enable and start Logstash
  systemd:
    name: logstash
    enabled: true
    state: started
```

```
- name: Start Logstash
  ansible.builtin.systemd:
    name: logstash
    enabled: yes
    state: started
```

Analysis:

- **This main.yml for logstash mainly imports the elastic's GPG key with its rpm_key and creates a yum repository for the 8.x packages. In addition, it also installs logstash with the use of dnf, then enables and starts the service using systemd so it is able to run in the background.**

8) Running Ansible Playbook

```
PLAY RECAP *********************************************************************
192.168.56.102             : ok=5    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.103             : ok=9    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.104             : ok=9    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

masongsong@Masongsong:~/CPE212_Activity10$
```

Analysis:

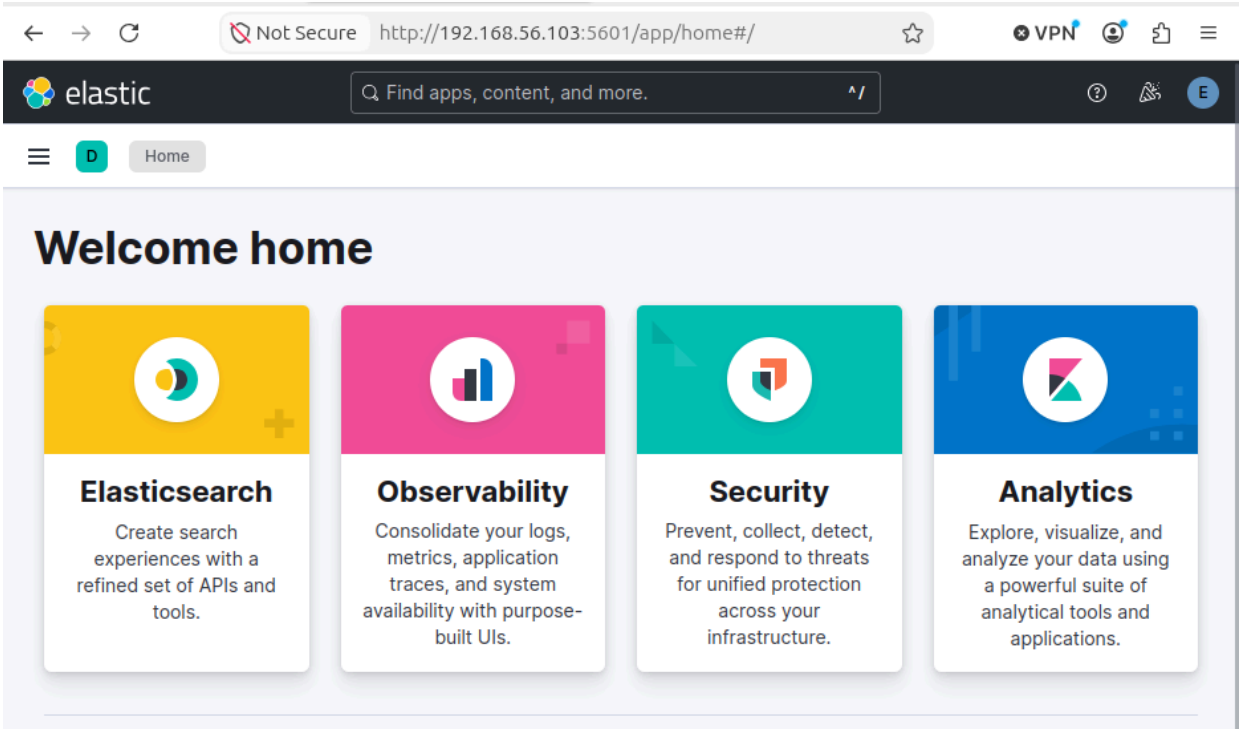
- **This shows that the playbook ran without errors, meaning everything is now working and elasticsearch, kibana, and logstash have successfully been installed on both the ubuntu and centos systems.**

9) Verification of Installed tools

KIBANA

```
masongsong@Masongsong:~$ sudo systemctl status kibana --no-pager
[sudo] password for masongsong:
● kibana.service - Kibana
     Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: en
abled)
     Active: active (running) since Mon 2026-10-05 09:33:50 UTC; 1h 30min ago
       Docs: https://www.elastic.co
   Main PID: 1097 (MainThread)
      Tasks: 11 (limit: 1432)
     Memory: 549.6M (peak: 885.7M)
        CPU: 1min 57.702s
     CGroup: /system.slice/kibana.service
             └─1097 /usr/share/kibana/bin/../node/glibc-217/bin/node /usr/share…
```

ELASTIC



LOGSTASH

```
masongsong@Masongsong:~/CPE212_Activity10$ ssh masong@192.168.56.102
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Mon Oct  5 23:10:03 2026 from 192.168.56.101
[masong@localhost ~]$ systemctl status logstash
● logstash.service - logstash
     Loaded: loaded (/usr/lib/systemd/system/logstash.service; enabled; preset:>
     Active: active (running) since Tue 2026-10-06 16:14:58 PST; 1min 24s ago
   Main PID: 769 (java)
      Tasks: 15 (limit: 10500)
     Memory: 541.3M (peak: 541.8M)
        CPU: 23.357s
     CGroup: /system.slice/logstash.service
             └─769 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -Djava.awt.he>

Oct 06 16:14:58 localhost systemd[1]: Started logstash.
Oct 06 16:14:59 localhost logstash[769]: Using bundled JDK: /usr/share/logstash>
```

Reflection:

What are the benefits of having a log monitoring tool?

- **By having a log monitoring tool, we are able to collect logs from different servers and compile it into one place, so administrators are able to search and analyze the logs without the need for them to log into each machine using SSH.**

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

CONCLUSION:

I have learned a lot from this HOA, mainly how to setup ansible in a way that it is able to download 3 different log monitoring tools on 3 different VMs. It was quite hard to grasp the configuration that is needed in order for these tools to run, but after some time researching, I was able to pull it off and make it work. All in all, I still enjoyed this HOA even though it gave me quite a challenge.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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